

TechQuake – Visualizing Layoffs in the Technological Landscape

Choice of Design Prototypes

We want to help the U.S. workforce understand the tech layoff crisis and guide their career choices. We also want to provide economic planners with insights to develop effective tech layoff mitigation strategies. Hence, we designed five visualizations that will be integrated into our interactive dashboard.

Our first goal for this project is multi-layered: to analyze layoffs over time and relate them to company performance via share price trends post-pandemic. For the first part, we have constructed a dual-axis bar graph to facilitate monthly comparisons of tech layoffs and the companies involved. This allows users to effortlessly comprehend the volume of tech layoffs and its spread across companies, along with potential temporal patterns.

For the second part, we designed an interactive multi-view composition of a time series plot and a data table. This enables users to focus on publicly-traded companies and analyze their prices relating to COVID-19 and layoff dates. Consequently, users can discern the domino effects of the pandemic on a tech company's layoff decisions.

Next, we aim to identify the tech subsectors most prone to workforce adjustments. We decided on a heatmap illustrating the layoffs in these subindustries over time. This visualization provides a comprehensive overview of layoffs across all subsectors, helping users to discern which subsector experienced the most layoffs from 2022 – early 2024.

We also want to understand whether tech layoffs vary by funding stages. For this, we designed a compound figure combining side-by-side boxplots and a summary table. This approach makes it easy for users to grasp the distribution and variability of tech layoffs across all financing stages. The summary table complements the visual analysis by providing key statistics.

Our last goal is to identify the concentration of tech layoffs in the country. Here, we implemented a Shiny bar graph visualization that can be filtered by year and city. This intuitive design makes it straightforward for audiences to identify the cities with the highest tech layoffs by year or across all years.

Demonstration and Implementation Discussion

Our dual-axis bar chart displays the tech layoffs in 2022–2024 from two perspectives: employees and companies. We realized that transforming the data into a long format and utilizing ggplot2’s dodging would not work because of the different y-axes scales. Therefore, we overlaid two distinct bar plots to represent each metric and nudged them horizontally to emulate ggplot2’s dodging. While adding the second y-axis for total companies, we had to scale it accordingly to ensure this metric’s visibility. We also encoded each metric with different colors to make them easily distinguishable.

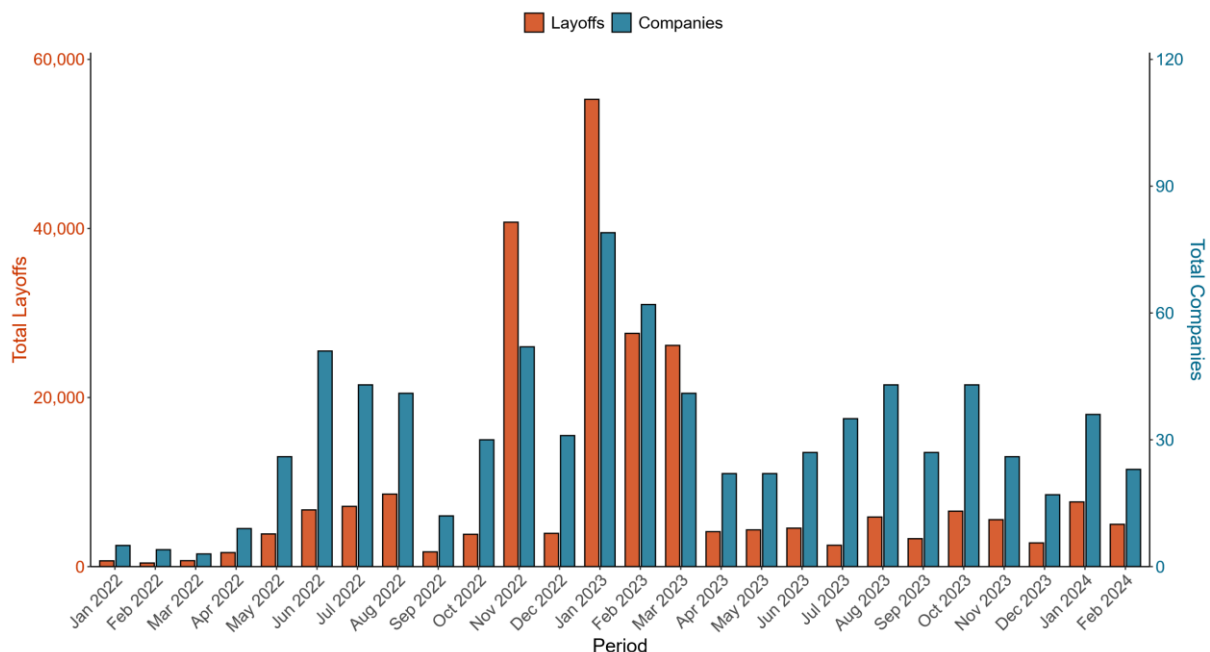


Figure 1: Tech Layoffs in 2022-2024.

The Shiny time series plot and data table shows COVID-19 impacts on companies' share prices and layoff decisions. The prices data was scraped from Yahoo Finance with Rvest. We allow users to select a publicly-traded company, in which the plot and table will be updated accordingly. We also highlighted parts of the plot to represent the phases of COVID-19 that various companies had gone through and how it has impacted their performance. This enables users to understand the starting point of a downturn for tech companies, leading to layoffs.

Our heatmap visualizes fluctuations in layoffs across subindustries over time. As part of the setup, we ensured that each subsector has an observation at each period, allowing us to populate each title easily. To highlight the relative differences of layoffs more effectively, we used a log scale of total layoffs to fill the titles and ordered the subindustries based on decreasing monthly layoffs. Thus, users can clearly make sense of which subindustries have the most layoffs.

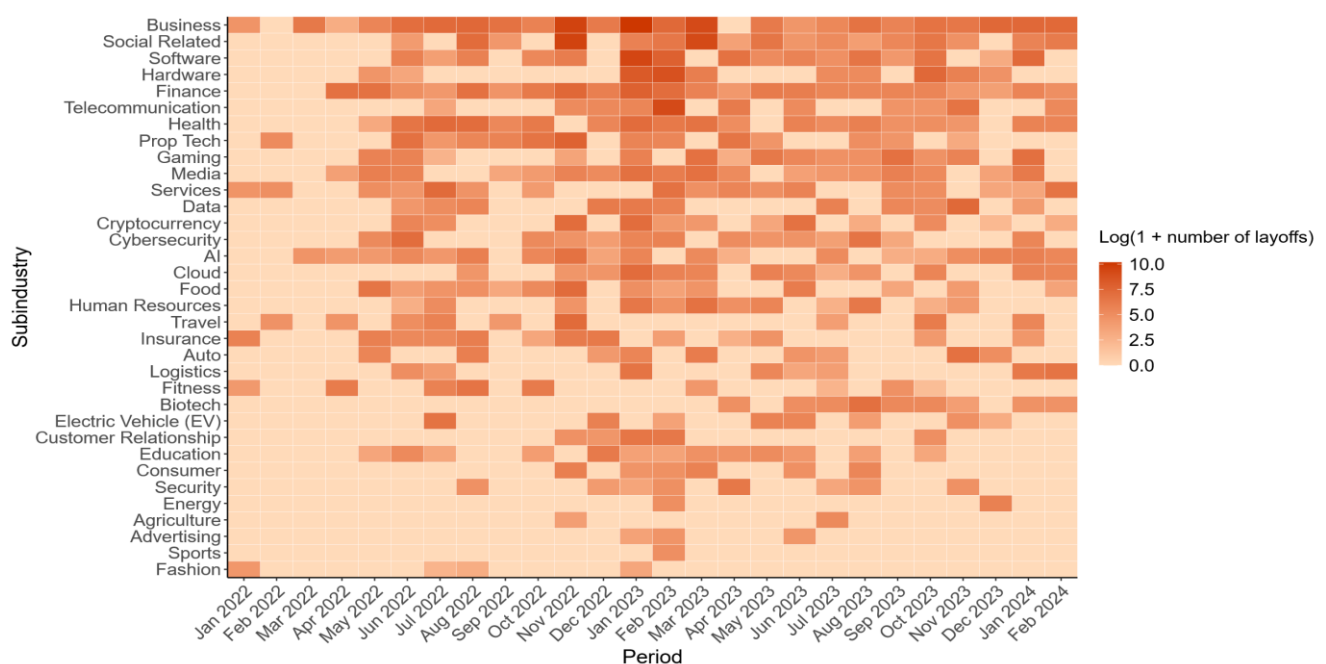


Figure 2: Tech Layoffs within Subindustries.

Our side-by-side boxplots and a summary table compound figure shows how layoffs are distributed across financing rounds of tech companies. We first filtered out the stages with five

or less observations to avoid displaying misleading distributions. We then transformed the boxplot's y-axis into a logarithmic scale to handle wide variances in layoff numbers. For the table, we included the mean layoffs and the five-number summary of each stage. To improve readability, we ordered the funding rounds by decreasing median number of layoffs. By leveraging both graphical and numerical representations, users such as economic planners can perform deeper analysis to plan company growth strategies.

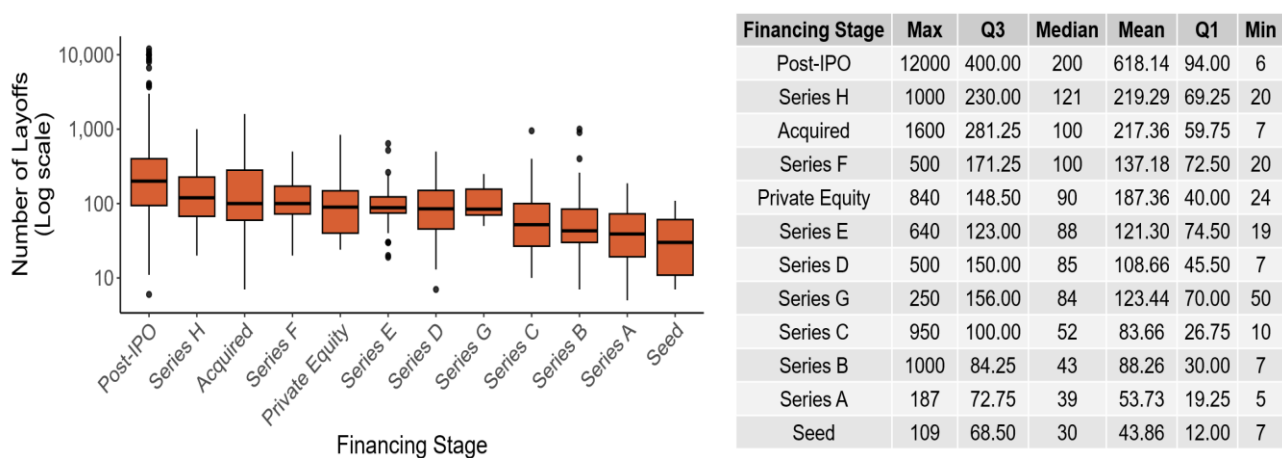


Figure 3: Tech Layoffs by Financing Stage.

The Shiny bar graph shows tech layoffs by year(s) and city/cities of one's choice. Since not all cities have layoff records for each year, we designed its available choices to be dependent on the year selection. When hovering over the bars, users can view the precise layoff figures for each city so that they have a greater sense of the magnitude of layoffs. To enhance clarity, we used a color gradient to indicate relative layoffs and reordered the bars by decreasing total layoffs.

Critical Evaluation

The dual-axis bar chart suffers from overcrowding due to our time measure and the presence of two bars per period. We contemplated graphing a line plot for its efficient use of space and easy comparisons between the layoffs and the number of companies involved, similar to the bar chart. However, we prefer our former design because the line plot tends to smooth when comparing layoffs by months, potentially causing data misrepresentation.

Focusing on our combined side-by-side boxplots and a summary table, we lost layoff information about certain financing stages with insufficient observations. We had to exclude them because their distributions would not be well represented with little data. Furthermore, including both visual and numerical representations might overwhelm certain audiences due to their information density. Because of this, we considered plotting a bar graph to visualize the average number of layoffs by financing rounds. However, we felt that it did not achieve our intended goal of finding variance in layoff distributions among stages effectively. Thus, we settled with the multi-view composition design.

Turning to our Shiny bar graph, users could not see the exact location of where tech layoffs are occurring. Perhaps, an interactive map would be a more practical visualization. However, the absence of geographical coordinates in our data made this implementation challenging. Still, we believe our Shiny bar graph does the job by offering users the ability to compare and analyze tech layoffs precisely across cities and gather where they are most concentrated.

In our final interface, we intend to divide the visualizations into multiple tabs: a general and a company-specific view. The latter houses our interactive multi-view composition of time series plot and data table, whereas the former includes the other visualizations. Our analysis of COVID-19 on a company's market performance and layoff decisions is much more focused, hence we feel it is necessary to include a separate section for this visualization. We also plan to allow interactivity among the general plots. For instance, by brushing over the dual-axis bar

chart to select specific tech layoff periods, visualizations such as the Shiny bar graph no longer need its own year selection for it to reflect a certain timeframe. We believe that being able to examine the entire tech layoff period and specific intervals enhance users' understanding of this deepening crisis.